TEACHER GUIDE

WJEC Eduqas GCSE (9-1) in Business (C510QS)

# Investment appraisal

Finance

**Revenue and costs: average rate of return**



A revision lesson. Pupils run a car dealership in firms of four and revise investment appraisal on their own deals. Seat each firm together. Each firm answers the topic tasks together.

## The lesson

**45-minute setting**

| Setup | Round 1 | Round 2 | Round 3 | Final questions |
|---|---|---|---|---|

### EVERY ROUND

**Firm decisions**
3 min

→

**Deals**
4½–5 min

→

**Topic tasks**
4 min

→

**B2B deal**
2½ min
from Round 2

### LESSON SETTINGS

| 30 min | 45 min | 60 min | 75 min |
|---|---|---|---|

# What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



## CEO

Staff · Sabotage from Round 2



## Finance

Capital & Loans · Investments



## Sales & Marketing

Marketing



## Operations

Seasonal stock

# Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

## 1 Lifetime profit

Round 1

A new valeting bay.

**Total profit over its life?**

| | |
|---|---|
| Cost to set up | **£45,000** |
| Returns, each year | **£14,500** |
| It lasts | **5 years** |

**ANSWER**

**£27,500**

5 × £14,500 − £45,000 = £27,500

**Why:** It earns back its cost and £27,500 more.

## 2 What ARR misses

Round 1

You compare two investments by ARR.

**Which two things does ARR not show?**

- **A** The cost of the investment
- **B** When the money comes back
- **C** The average profit a year
- **D** The risk that returns fall short
- **E** Profit as a % of the cost

**ANSWER**

**✓ When the money comes back**
**✓ The risk that returns fall short**

**Why:** ARR averages the profit; it says nothing of timing or risk.

## 3 ARR steps in order

Round 2

You work out the ARR of a new ramp.

**In what order do you work it out?**

- **A** Divide by the years
- **B** Add up all the returns
- **C** Divide by the cost, × 100
- **D** Take off the cost: total profit

**ANSWER**

**1. Add up all the returns**
**2. Take off the cost: total profit**
**3. Divide by the years**
**4. Divide by the cost, × 100**

**Why:** Total profit, then profit a year, then as a % of the cost.

# Topic tasks

## 4 Valeting bay ARR

Round 2

A new valeting bay.

**Average rate of return (ARR), as a %?**

| | |
|---|---|
| Cost to set up | **£60,000** |
| Profit over its life | **£84,000** |
| It lasts | **5 years** |

**ANSWER**

**28%**

£84,000 ÷ 5 = £16,800; ÷ £60,000 × 100 = 28%

**Why:** It earns 28p a year for every £1 put in.

## 5 ARR below loan rate

Round 3

A ramp would have an ARR of 6%. You would borrow the money at 8% a year.

**Should you invest, and why?**

**A** No

**B** Yes

**ANSWER**

✔ **No**

**Why:** A 6% return does not cover an 8% cost of borrowing.

## 6 Better option ARR

Round 3

Which one pays better?

**ARR of the better option, as a %?**

| | |
|---|---|
| Option A: cost | **£75,000** |
| Option A: profit each year | **£9,000** |
| Option B: cost | **£55,000** |
| Option B: profit each year | **£13,750** |

**ANSWER**

**25%**

A 12%; B 25%; better 25%

**Why:** Compare the rates, not the pounds of profit.

# Final questions

Each pupil answers 6 final questions on investment appraisal. They come from these 10 questions.

Then each pupil writes one short answer: “Explain why ARR is compared with the interest rate.”

1. A workshop upgrade costs £60,000 and is forecast to make total profit of £15,000 over three years. What is its average rate of return?
   ✔ **8.3%**

2. A project is forecast to make £3,000, £5,000 and £4,000 profit over three years. What maximum investment gives a 20% average rate of return?
   ✔ **£20,000**

3. Upgrade A: cost £50,000, £12,000 profit over three years. Upgrade B: cost £40,000, £10,800 over three years. Which has the higher average rate of return, and by how much?
   ✔ **Upgrade B, by 1 percentage point**

4. Two investments have similar average rate of return. What additional factor should the Finance lead consider?
   ✔ **Risk and when the cash comes back**

5. A new MOT bay costs £40,000 and should make £24,000 total profit over four years. What is the average rate of return?
   ✔ **15%**

6. A paint booth costs £80,000 and is expected to make £10,000 profit every year. What is its average rate of return?
   ✔ **12.5%**

7. Project A costs £30,000 and makes £9,000 total profit over three years. Project B costs £50,000 and makes £12,500 over five years. Which has the higher average rate of return?
   ✔ **Project A, at 10%**

8. What does the average rate of return tell a dealer about an investment?
   ✔ **Average yearly profit as a percentage of the cost**

9. A dealer spends £40,000 on a car wash. Over 4 years it makes a total profit of £24,000, after taking off the cost. What is the ARR?
   ✔ **15%**

10. A dealer can buy a car wash with an ARR of 12% or a new ramp with an ARR of 8%. On ARR alone, which should it choose?